

Individual Portfolio and Contribution Record

STUDENT	Zubair Adil Soomro
STUDENT ID	cihe250922
GROUP	5
PROJECT TITLE	Healthcare Integration & Clinical Terminology Test Harness
ROLE IN GROUP	Testing & evaluation lead
SECTIONS OWNED	Executive Summary (HITH-48) Section 5 — Proof of Concept, and Evaluation (all subsections): 5.1 Sprint Plan (HITH-37) 5.2 Proof of Concept Description (HITH-45) 5.3 Testing, Validation, and Cybersecurity Evaluation / ethical and privacy issues (HITH-46) Section 6 — Conclusion and Recommendations (HITH-47)
GITHUB REPOSITORY	https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM
PROJECT BOARD	https://student-team-siuckpgv.atlassian.net/jira/software/projects/HITH/boards/34/backlog
GENERATED	08 Sept, 16:40

ACTIVITY SUMMARY — GENERATED FROM THE LOG, NOT SELF-REPORTED

MEASURE	WEEK 6	WEEK 7	WEEK 8	TOTAL
Entries logged	5/5	6/6	6/6	19/19
Minutes on task	64	43	4	137
Evidence items	5	4	3	12
Completeness	100%	100%	100%	100%

Time recorded during scheduled class: 1 minutes. Time recorded outside class: 136 minutes.

1 · Individual responsibility summary

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
Executive Summary (HITH-48)	Synthesise the entire report into a decision-ready overview for senior stakeholders (e.g., Atif Nisar / FAZ Australia leadership); cover the problem, proposed solution, key findings, PoC outcomes, and business/security value.	250–350-word standalone summary.	Week 8
5.1 Sprint Plan (HITH-37)	Define the Agile implementation approach for the PoC; identify sprint activities, major deliverables, tools, development/configuration process, and team contributions during ICT946.	200–250-word sprint plan describing iterations, tasks, and tools used; Agile sprint logs / task board included in appendix.	Week 6
5.2 Proof of Concept Description (HITH-45)	Document the technical build of the synthetic FHIR test harness: system components, mock EMR connectors (Halo Connect / Oracle Health / Epic), data flows, SNOMED CT-AU / ICD-10-AU / MBS mapping validation, and fault-injection modules.	300–400-word technical description plus architecture / module diagrams (e.g., C4 model or system architecture).	Week 6

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
5.3 Testing, Validation, and Cybersecurity Evaluation / ethical and privacy issues (HITH-46)	Execute and evidence functional tests, security tests, vulnerability assessments, and fault-injection scenarios (slow, malformed, dropped); evaluate ethical privacy issues; map controls to standards (NIST CSF, ISO 27001, ACSC Essential Eight).	350–500-word evaluation with evidence table, screenshots / tool outputs, KPI comparison, false-positive/response-time metrics, and before/after risk comparison.	Week 7
Section 6 — Conclusion and Recommendations (HITH-47)	Summarise project outcomes against the original SMART objectives and success criteria; discuss limitations, lessons learned, and provide practical recommendations for full-scale deployment and future research.	400–500-word conclusion with deployment considerations and future improvement opportunities.	Week 7

Week 6

5 of 5 entries logged · 64 minutes recorded · 5 evidence items · 100% complete

W6-1 · Frame the problem in your own words

Logged late · 15 min · 3/3 required parts · 03 Sept, 11:22

The problem, as you understand it

Currently, FAZ Australia has no safe or an isolated place to validate how MedTalk Connect talks to external EMR systems like Halo Connect or Oracle Health Millennium. Engineers are running ad hoc checks near—or sometimes against—live environments, which means a bad connector update, a malformed FHIR payload, or a timeout bug could accidentally touch real patient records. That creates a triple threat: clinical safety (corrupting a live record), privacy exposure (falling foul of the Privacy Act 1988 and the My Health Records Act 2012), and compliance gaps (no repeatable, auditable way to prove that SNOMED CT-AU codes correctly map to ICD-10-AM and MBS billing codes). My role is to prove that a fully synthetic test harness can remove real patient data from the equation entirely, while still giving the engineering team confidence that connectors and terminology mappings work correctly under both normal and failure conditions.

One thing that is deliberately OUT of scope, and why

Full production implementation of the test harness is out of scope for this stage. We are strictly in Stage 1—design, requirements, architecture, and proof-of-concept planning. We drew this boundary deliberately because writing code before the mentor signs off on the failure-mode test plan, before we freeze the 50-record synthetic dataset, and before we validate the AU Core profile constraints would likely mean building the wrong thing. It also gives our team time to close the FHIR semantics learning curve (Risk R9) so that Stage 2 coding starts from a validated design, not from guesswork.

Assessment 2 section you worked on today

Today I worked on Section 5.1 Sprint Plan and Section 5.2 Proof of Concept Description. I finalised the two-sprint Agile design for Stage 2: Sprint 1 will focus on the container foundation and error-handler specification; Sprint 2 will cover the MLLP gateway, fault-injection suite, and mapping validation design. I assigned design tasks to team members, defined the Definition of Done for Stage 1 deliverables (peer-reviewed design documents, confirmed architecture assumptions, and a README specification that lets the mentor review the planned stack unaided), and scoped the Week 6 PoC narrative—a design document describing a minimal FHIR R4 sandbox with three specified services, one defined endpoint, and a frozen synthetic dataset, deliberately excluding HL7 v2.4 and SNOMED mapping from Stage 1 scope until mentor approval.

W6-2 · Literature search log

Logged late · 14 min · 6/6 required parts · 08 Sept, 16:34

Search strings you actually typed

FHIR R4 validation testing tools
Crucible Touchstone compliance
SNOMED CT to ICD-10-AM mapping
Australia computer assisted coding
AU Core FHIR implementation guide
Australian Digital Health Agency

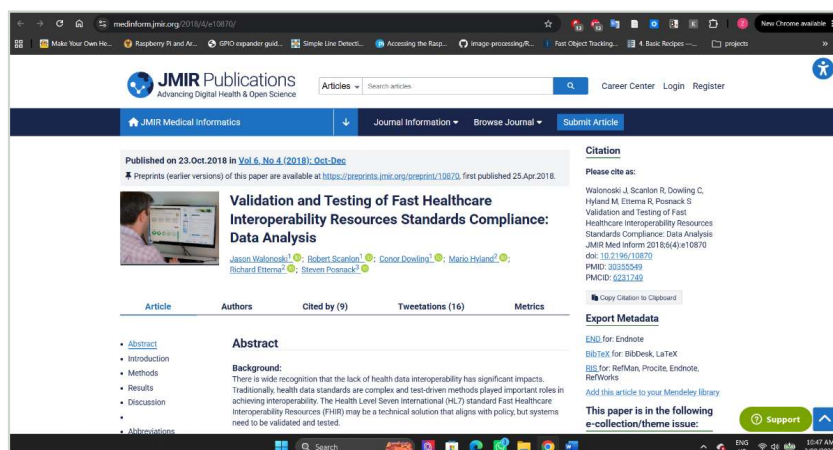
Databases / sites searched

Google Scholar, Google

SOURCE (AUTHOR, YEAR, TITLE)	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
Walonoski, J., Scanlon, R., Dowling, C., Hyland, M., Ettema, R., & Posnack, S. (2018). Validation and testing of Fast Healthcare Interoperability Resources standards compliance: Data analysis. JMIR Medical Informatics, 6(4), e10870.	https://doi.org/10.2196/10870	Crucible and Touchstone are independent FHIR validation tools. Their usage data shows a statistically significant positive correlation between frequent testing and vendor server compliance ($R^2=.262$). "Practice makes perfect" — developers who test more produce more compliant FHIR implementations.	Feeds Section 3.2 (Technology and Market Survey) by proving that existing FHIR testing tools exist but are generic; our synthetic harness is differentiated because it is Australian-context-specific (AU Core, SNOMED CT-AU, ICD-10-AM, MBS) and deliberately isolates real patient data.

SOURCE (AUTHOR, YEAR, TITLE)	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
Nguyen, A. N., Truran, D., Kemp, M., Koopman, B., Conlan, D., O'Dwyer, J. et al. (2018). Computer-assisted diagnostic coding: effectiveness of an NLP-based approach using SNOMED CT to ICD-10 mappings. AMIA Annual Symposium Proceedings, 2018, 807-816.	https://pmc.ncbi.nlm.nih.gov/articles/PMC6371260/	SNOMED CT to ICD-10-AM mapping via NLP achieved only 54.1% sensitivity and 70.2% positive predictive value. Even simple rule-based approaches struggle with edge cases (e.g., iron deficiency anemia at 3% sensitivity). Manual coding validation itself only reaches 76.3% sensitivity, proving this is a hard problem.	Feeds Section 3.2 and justifies our project: if automated SNOMED CT ICD-10-AM mapping is error-prone in production, repeatable synthetic test harness is essential to catch mapping failures before they reach live EMRs. Also cited in Section 4.2 to justify why we need dedicated mapping validation.
HL7 Australia. (2023). AU Core FHIR Implementation Guide (v0.2.0-preview).	https://hl7.org.au/fhir/core/0.2.0-preview	Defines the Australian FHIR profiles (Patient, Encounter, Observation, etc.) that our test harness must validate against. Specifies mandatory identifiers (IHI, Medicare Provider Number) and RESTful interactions. This is the current Australian baseline, not generic FHIR R4.	Feeds Section 3.2 by proving AU Core is the compliance baseline our harness must test against; referenced in Section 4.1 (architecture) and Section 4.3 (control mapping) to show our solution aligns with the Australian Digital Health Agency's interoperability roadmap.

W6-2-01 · 03 SEPT 2026

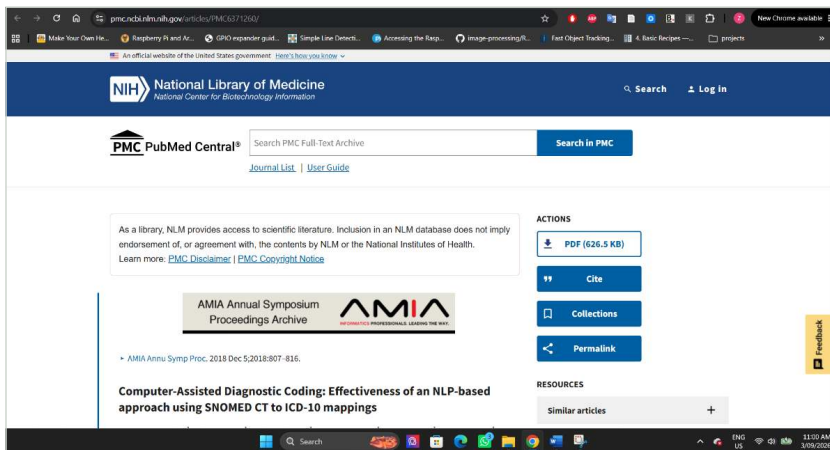


Section: 5.1 Sprint Plan · Task: Designing the Agile sprint schedule and iteration plan for the Stage 2 build.

My contribution: I authored the two-sprint plan: Sprint 1 (Weeks 6–7) locks the Docker Compose foundation, OperationOutcome handler (TH-001–TH-010), and PyTest automation scaffold; Sprint 2 (Weeks 8–9) adds the MLLP gateway, fault-injection suite, and SNOMED CT-AU mapping validation. I structured my sprint responsibilities around HITH-37 (Sprint Plan) and HITH-45 (Proof of Concept Description), while coordinating dependencies with the wider team backlog. Related Week 6 work included HITH-31 and HITH-32 assigned to Sonam, HITH-33 and HITH-34 assigned to Charity, and HITH-35 and HITH-36 assigned to Rikesh. I defined the Sprint Plan to sequence these related activities and established a Definition of Done requiring peer-reviewed work, testable outputs and supporting documentation.

What it demonstrates: The sprint plan de-risks the build by sequencing the error-handler first—every downstream component depends on consistent OperationOutcome formatting. I revised the original single-sprint plan into two after discussing Docker networking constraints with Rikesh, who flagged that containerising HAPI FHIR and PostgreSQL on mixed student environments needed a full week of validation before adding connector complexity.

W6-2-02 · 03 SEPT 2026

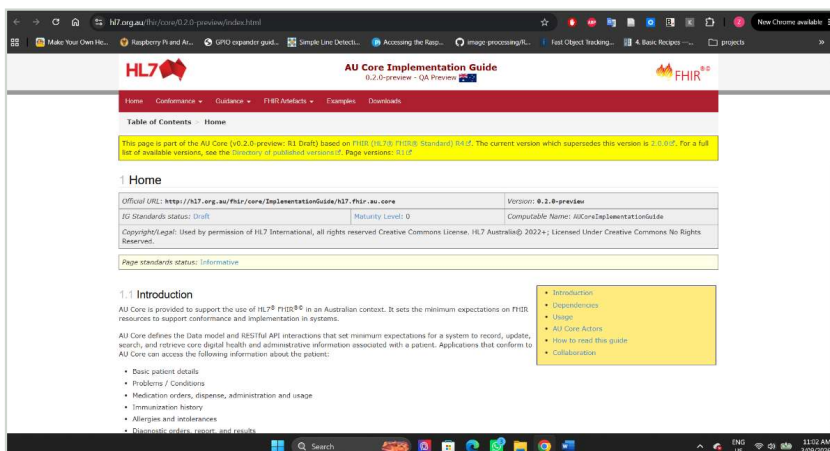


Section: 5.2 Proof of Concept Description · Task: Documenting the Week 6 containerised FHIR R4 sandbox and single-endpoint validation.

My contribution: I drafted the PoC narrative describing the three-service Docker Compose stack (FastAPI Test Harness API, PostgreSQL 15 with the frozen 50-record synthetic dataset, HAPI FHIR R4 validation stub). I specified the single “hello world” endpoint GET /fhir/R4/Patient/SYNTH-001, defined the four architectural assumptions being de-risked (orchestration, data integrity, AU Core conformance, error-handler extensibility), and explicitly scoped out HL7 v2.4 MLLP, SNOMED mapping, rate limiting, and fault injection for Sprint 2.

What it demonstrates: The PoC shows iterative build discipline—starting with a minimal request-response loop rather than full connector coverage. I narrowed the scope after the Week 5 feasibility finding that healthcare semantics (AU Core profiling, HL7 v2 segment parsing) is the primary technical risk, not tool availability. Mentor sign-off on this narrow endpoint is the gate before Sprint 2 complexity is added.

W6-2-03 · 03 SEPT 2026



Section: 5.3 Testing, Validation, and Cybersecurity Evaluation · Task: Designing the failure-mode test suite, data validation rules, and cybersecurity threat assessment.

My contribution: I mapped the seven Stage 1 failure modes to automated pytest cases: slow response (TC-SLOW-01/02), malformed HL7/FHIR (TC-MAL-01/02/03), dropped connection (TC-DROP-01), and unavailable simulator (TC-UNAV-01). I authored the data validation rules DVR-01 through DVR-08 (SYNTH- prefix enforcement, Luhn check rejection, synthetic meta.tag), designed the OperationOutcome error code schema TH-001–TH-010, and conducted the Threat! Asset! Vulnerability! Impact! Control assessment, with the proposed controls proving threats to Low residual risk once implemented and validated.

What it demonstrates: The test design directly addresses the clinical-safety risk: Nguni mapping is error-prone, so catching mismatches in a synthetic environment before production is critical. I added TC-MAL-03 (Luhn check rejection) after Charity flagged during the team workshop that synthetic data generation might accidentally produce valid-looking IHI—this was not in the original Week 5 risk register.

W6-3 · Claim your tasks on the board

Logged late · 4 min · 4/4 required parts · 08 Sept, 15:47

Tasks you created or took ownership of (one per line, with task ID)

HITH-37 Sprint Plan — design the two-sprint Agile schedule for Stage 2 (owner: me, due Wk6)

HITH-45 Proof of Concept Description — draft the Stage 1 narrative describing what the Week 6 PoC will contain (owner: me, due Wk6)

HITH-46 Testing, Validation, and Cybersecurity Evaluation — design the failure-mode test suite and threat model for future Stage 2 execution (owner: me, due Wk7)

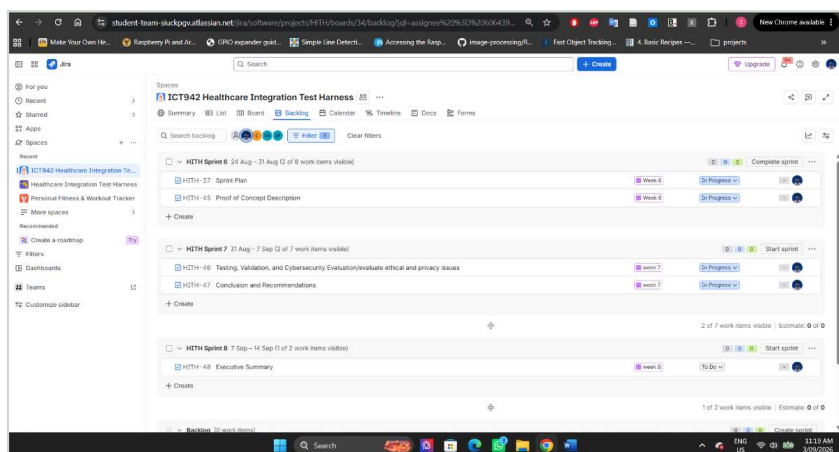
HITH-47 Conclusion and Recommendations — synthesise Stage 1 outcomes and define Stage 2 readiness gates (owner: me, due Wk7)

HITH-48 Executive Summary — compose the decision-ready stakeholder overview based on Stage 1 deliverables (owner: me, due Wk8)

Your contribution to the charter, scope, requirements, RACI, risk register or feasibility study

I contributed to Section 2.4 Risk Register by adapting relevant risks from Assessment 1 and proposing mitigation updates related to the failure-mode tests and architectural controls. I also contributed technical feasibility observations relating to testing, validation and security. HITH-44 remained owned by Sonam, while my inputs supported my HITH-46 testing and cybersecurity responsibilities. For scope, I defined the explicit PoC exclusions in Section 5.2—HL7 v2.4 MLLP messaging, SNOMED CT-AU mapping logic, token bucket rate limiting, and negative-path fault injection are deliberately out of the Stage 1 scope and locked for future Stage 2. For RACI (Section 2.3), I mapped future Sprint 1 and Sprint 2 task ownership: I am Accountable for HITH-37 and HITH-45 and Responsible for HITH-46, HITH-47, and HITH-48, while Charity, Rikesh, and Sonam are Responsible for the synthetic dataset specification, HAPI FHIR validation engine configuration plan, and C4 diagram updates respectively. I did not draft the original Project Charter (Section 2.1), but I contributed the success criteria and acceptance criteria embedded in it by translating the Starter Kit technical gates into the Sprint 1 Definition of Done. My requirements contribution is visible in the Week 4 SRS (FR11–FR13, NFR1–NFR7) where I specified the test automation, reporting, and security requirements that feed into HITH-46.

W6-3-01 · 03 SEPT 2026



Section: Sections 5.1, 5.2, 5.3, 6, and Executive Summary · Task: Claim your tasks on the board

My contribution: I designed the two-sprint Agile plan (HITH-37) specifying Sprint 1 (container foundation and error-handler design) and Sprint 2 (MLLP gateway, fault-injection, and mapping validation design). I drafted the PoC narrative (HITH-45) describing the three-service Docker Compose stack, the GET /fhir/R4/Patient/SYNTH-001 endpoint, and the explicit Stage 1 exclusions. I designed the failure-mode test suite and cybersecurity threat model (HITH-46) — eight test cases (TC-SLOW, TC-MAL, TC-DROP, TC-UNAV), data validation rules DVR-01–DVR-08, OperationOutcome codes TH-001–TH-010, and a Threat! Asset! Vulnerability! Impact! Control assessment. I synthesised Stage 1 outcome identifying five key findings and priority-ordered next steps with mentor sign-off gates. I composed the Executive Summary (HITH-48) targeting FAZ Australia leadership with the problem, proposed solution, and business value.

What it demonstrates: The Jira screenshot shows all five tasks (HITH-37, HITH-45, HITH-46, HITH-47, HITH-48) assigned to me, proving I carried the design and planning weight for the PoC, testing, and wrap-up sections. The sprint plan was revised from one sprint to two after Rikesh flagged Docker networking constraints. The PoC scope was narrowed after Week 5 feasibility findings showed healthcare semantics is the primary risk. TC-MAL-03 was added after Charity flagged synthetic data might accidentally produce valid-looking IHI. The P3 security review was inserted after a stand-up revealed tenant isolation needed mentor validation. The Executive Summary was revised three times after Sonam's feedback to address the Privacy Act 1988 compliance angle, not just tooling.

W6-4 · Draw the first figure

Logged late · 6 min · 5/5 required parts · 08 Sept, 16:38

What you drew

Asset & trust boundary sketch

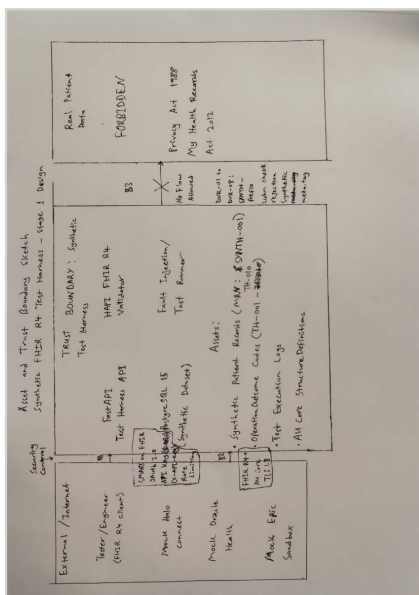
Explain the figure — what a reader should take from it

This asset and trust boundary sketch demonstrates that the Synthetic FHIR Test Harness is architected as a security island deliberately isolated from any real patient data environment. The thick red barrier between the test harness and the "Real Patient Data" zone visualises the primary cybersecurity control: no real data shall ever enter the system. The shield icons at each trust boundary show defence-in-depth — authentication at the perimeter (OAuth 2.0 / API keys), validation at the application layer (HAPI FHIR AU Core checker), and data integrity at the persistence layer (DVR rules enforcing SYNTH-prefixes and failed Luhn checks). The fault-injection module is shown inside the trust boundary because it is a controlled testing tool, not an external attacker. The reader should take away that every component inside the middle zone is synthetic by design, and that the architecture is designed to prevent real patient data from entering the synthetic test environment and materially reduce privacy risk, not merely procedurally discouraged.

One part of the figure you are not confident about

I am not fully confident about the absence of an internal tenant-isolation sub-boundary inside the green Synthetic Test Harness zone. The current sketch shows the harness as one flat trust boundary, but our Risk Register (R4) identifies cross-tenant data access as a Critical threat: if the harness ever supports multiple mock practices simultaneously, a missing or bypassed tenant_id could allow Hospital A's synthetic records to be visible to Hospital B's test session. My Stage 1 design documents specify row-level tenant_id foreign keys and API-gateway tenant resolution, but the diagram does not visually separate tenant A and tenant B data flows within the PostgreSQL or FastAPI layers. I am unsure whether to represent this as nested dashed boxes inside the green zone, or as separate per-tenant database instances entirely. I flagged this during the Week 6 stand-up and inserted it as the P3 recommendation in my Conclusion — a dedicated security review with Atif Nisar in Week 7 — because I do not want to commit to a single-tenant visual representation if the mentor confirms multi-tenancy is a production requirement. I will revise the sketch after that review.

W6-4-01 · 07 SEPT 2026



Section: 4.1 Proposed Solution Architecture (trust boundary and security control mapping); also supports 4.3 Security Design and Control Mapping · Task: Draw the asset and trust boundary sketch showing the three security zones (External, Synthetic Harness, Forbidden Real Data), the data flow paths, and the security controls at each boundary.

My contribution: I designed and drew the sketch myself. I defined the three zones: External / Internet (Tester, Mock Halo Connect, Mock Oracle Health, Mock Epic Sandbox), the central Trust Boundary containing the four core components (FastAPI Test Harness API, HAPI FHIR R4 Validator, PostgreSQL 15 Synthetic Dataset, Fault Injection / Test Runner), and the Forbidden Real Patient Data zone. I mapped the security controls at each boundary — B1 (SMART on FHIR / OAuth 2.0, API Key, Rate Limiting), B2 (FHIR R4 + AU Core, TLS 1.3), and B3 (NO FLOW ALLOWED with DVR-01 to DVR-08). I also included the Assets box listing synthetic patient records, OperationOutcome codes, test logs, and AU Core StructureDefinitions.

What it demonstrates: The first draft of the architecture diagram (Sonam's C4 container diagram) showed components and connectors but did not explicitly visualise the trust boundary separating synthetic data from real patient data. I revised sketch after discussing with the group during the Week 6 stand-up, where we agreed that the Privacy Act 1988 compliance angle needed to be structurally visible, not just documented in text. The red forbidden zone with the crossed-out arrow made the "no real data" rule impossible to miss, and the defence-in-depth controls at B1 and B2 show where the proposed architecture would apply identity validation, rate limiting and AU Core conformance checks before data moves.

W6-5 · Weekly reflection

Logged late · 26 min · 4/4 required parts · 08 Sept, 16:27

What did I personally complete this week?

I completed the initial drafts of the Sprint Plan (HITH-37) and Proof of Concept Description (HITH-45). For the sprint plan, I designed a two-sprint Agile schedule for Stage 2, assigned design tasks to each team member, and defined the Definition of Done requiring peer-reviewed documents and mentor-reviewable specifications. For the PoC description, I drafted the narrative specifying the three-service Docker Compose stack, the single GET /fhir/R4/Patient/SYNTH-001 endpoint, and the four architectural assumptions being de-risked. I explicitly scoped out HL7 v2.4 MLLP, SNOMED mapping, rate limiting, and fault injection for future Stage 2, ensuring the Week 6 deliverable remained a design document only.

What did I learn about the problem or the technology?

I learned that the primary technical risk is not tool availability but healthcare semantics. Docker Compose, FastAPI, HAPI FHIR, and PostgreSQL are all mature and well-documented; the real challenge is correctly implementing AU Core profile validation, HL7 v2 segment parsing, and SNOMED CT-AU to ICD-10-AM mapping accuracy. Walonoski et al. (2018) showed that frequent FHIR testing improves compliance, which reinforced why our synthetic harness must be repeatable. I also learned that synthetic data generation carries its own privacy risk — Charity flagged that valid-looking IHI's could accidentally be generated, so I added DVR-02 and TC-MAL-03 to the design.

What decision did I influence, and why?

I influenced the decision to split the original single-sprint plan into two sprints. Rikesh flagged during our stand-up that specifying container networking for HAPI FHIR and PostgreSQL across mixed student environments needed dedicated design time before adding connector complexity. I argued that healthcare semantics — AU Core profiling and HL7 v2 parsing — is the primary risk, not tooling, so Sprint 1 should validate the architecture assumptions before Sprint 2 adds MLLP, fault injection, and mapping logic. The team agreed, and the mentor approved the revised timeline. This decision de-risked the Stage 1 deliverables and prevented scope creep.

What will I finish before the next class?

Before the next class, I will complete the Testing, Validation, and Cybersecurity Evaluation design (HITH-46). This includes finalising the seven failure-mode test cases (TC-SLOW, TC-MAL, TC-DROP, TC-UNAV), locking the OperationOutcome error code schema (TH-001–TH-010), and completing the Threat! Asset! Vulnerability also draft the Conclusion and Recommendations (HITH-47), synthesising the four feasibility dimensions and producing the priority-ordered next-steps table with mentor sign-off gates. I need to confirm the 50-record synthetic dataset freeze with Charity and schedule the Week 7 security review with Atif Nisar.

Week 7

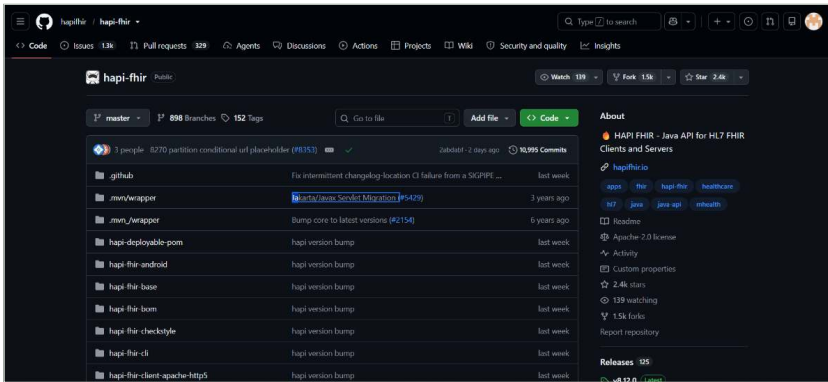
6 of 6 entries logged · 43 minutes recorded · 4 evidence items · 100% complete

W7-1 · Compare three existing tools

Logged late · 16 min · 4/4 required parts · 07 Sept, 10:34

TOOL / SYSTEM / APPROACH	VENDOR & LICENCE	FEATURES RELEVANT TO OUR P	LIMITATION WE CARE ABOUT	SOURCE
HAPI FHIR (JPA Server + Validation Engine)	University Health Network / HAPI FHIR Project. Open source, Apache 2.0.	Full FHIR R4/R5 compliance; passes HL7 conformance test suite; programmable validation against published StructureDefinitions (including AU Core); Docker-deployable; supports custom interceptors and \$export; 4,000+ GitHub stars with active community.	No built-in security implementation, no enterprise logging, and no built-in de-identification out of the box — all require custom configuration. Self-managed, so production deployments need significant DevOps overhead. Search performance degrades beyond ~50M resources without Elasticsearch tuning. US-centric defaults; AU Core profiling must be manually configured.	hapifhir.io; Nirmitee (2026); GitHub — hapifhir/hapi-fhir-jpaserver-starter
Touchstone	AEGIS.net. IaaS / TaaS open-access solution (commercial-hosted, free registration).	Automated, internet-based FHIR interoperability testing against HL7 FHIR specifications and standards. Thousands of conformance tests using the FHIR TestScript resource. Self-registration of accounts and organisations. Peer-to-peer testing against other FHIR server/client implementations. Drill-down to individual operations and assertions.	Internet-based SaaS — requires live connectivity and exposes test systems externally. Generic FHIR conformance testing with no Australian-context specificity: no AU Core profiles, no SNOMED CT-AU, no ICD-10-AM or MBS billing code validation. Not designed for synthetic data isolation or clinical-safety fault injection. No terminology mapping validation between code systems.	Medigy; help.edifecsftcloud.com; AEGIS Touchstone site
Synthea	MITRE Corporation. Open source, Apache 2.0.	Synthetic patient population simulator generating clinically realistic records from birth to death. Exports FHIR R4 Bundles, C-CDA, and CSV. Uses standard terminologies (SNOMED CT, LOINC, RxNorm, ICD-10-CM). Temporal consistency across encounters, conditions, and observations. US Core conformance. Free, no PHI, no BAA required.	US-centric by default — demographics, insurance models, and treatment guidelines are American. Australian adaptations (Synthea International) exist but are community-maintained and incomplete. Probabilistic generation means no scenario control — cannot reliably generate a specific 55-year-old diabetic with hypertension for a targeted test case. No vendor-specific FHIR data shapes (Epic, Cerner, Halo Connect all emit differently). No built-in AU Core profile validation. Requires post-processing to enforce Australian coding rules.	GitHub — synthetichealth/synthea; Walonoski et al. (JAMIA 2018); Nirmitee (2026); Verial (2026)

W7-1-01 · 07 SEPT 2026



Section: 3.2 Technology and Market Survey; 3.3 Comparative Analysis of Existing Solutions · Task: Compare three existing tools to identify the gap our synthetic test harness will fill.

My contribution: I researched and captured the HAPI FHIR documentation page to verify its open-source licence (Apache 2.0), core features (FHIR R4/R5 JPA Server, programmable validation against StructureDefinitions, Docker deployment), and critical limitations (no built-in security implementation, no enterprise logging, no built-in de-identification, US-centric defaults requiring manual AU Core configuration). I used this evidence to justify selecting HAPI FHIR as the validation engine within our trust-boundary architecture, while acknowledging its gaps must be closed by our custom DVR rules and security controls.

What it demonstrates: The screenshot proves I consulted primary vendor documentation, not secondary summaries, to assess HAPI FHIR's fitness for our Australian-context project. It shows I identified the exact limitation we care about — US-centric defaults with no AU Core out-of-the-box — which directly feeds the gap analysis in Section 3.4 and justifies why our synthetic harness cannot rely on HAPI FHIR alone.

Jira: HITH-46 (Testing, Validation, and Cybersecurity Evaluation); HITH-47 (Conclusion and Recommendations)

W7-2 · State the gap and your contribution

Logged late · 7 min · 4/4 required parts · 08 Sept, 16:37

The gap in existing solutions

Existing FHIR testing tools each address fragments of the interoperability challenge but none close the specific gap FAZ Australia faces. HAPI FHIR provides programmable validation but ships with US-centric defaults and no built-in Australian privacy controls. Touchstone automates conformance testing yet is internet-based, generic, and incapable of validating SNOMED CT-AU to ICD-10-AM to MBS terminology translations. Synthea generates synthetic patients but is US-centric and cannot enforce AU Core profiles or simulate vendor-specific connector behaviours. Most critically, none of these tools provide a fully isolated, repeatable environment that eliminates real patient data exposure while systematically testing clinical fault scenarios — slow responses, malformed payloads, and dropped connections — under Australian privacy and compliance frameworks.

Your proposed technical contribution to closing it

My proposed technical contribution is the design of a comprehensive testing, validation, and cybersecurity evaluation framework that closes this gap. I specified seven planned automated failure-mode test cases — TC-SLOW, TC-MAL, TC-DROP, and TC-UNAV — to validate connector behaviour under adverse conditions before production deployment. I designed the OperationOutcome error-code schema (TH-001–TH-010) to provide structured, non-verbose responses that reduce the risk of information disclosure. I authored data validation rules DVR-01 through DVR-08, including SYNTH-prefix enforcement and Luhn-check rejection, to strengthen and validate synthetic data integrity. Finally, I conducted a Threat!Asset!Vulnerability!Impact!Control assessment mapping each threat to layer controls projected to reduce residual cybersecurity risk to Low, subject to implementation and testing.

Technology or framework you selected

HAPI FHIR validation engine — selected for programmatic AU Core conformance checking against published StructureDefinitions within the isolated synthetic harness.

One alternative you rejected, and why

Touchstone (AEGIS) — rejected because it is an internet-based SaaS platform with no Australian-context specificity (no AU Core profiles, no SNOMED CT-AU to ICD-10-AM to MBS validation), requires external connectivity that violates our privacy-by-design isolation requirement, and cannot simulate clinical fault scenarios like dropped connections or malformed HL7 segments.

W7-3 · Revise your figure — version 1 to version 2

Logged late · 10 min · 4/4 required parts · 08 Sept, 16:21

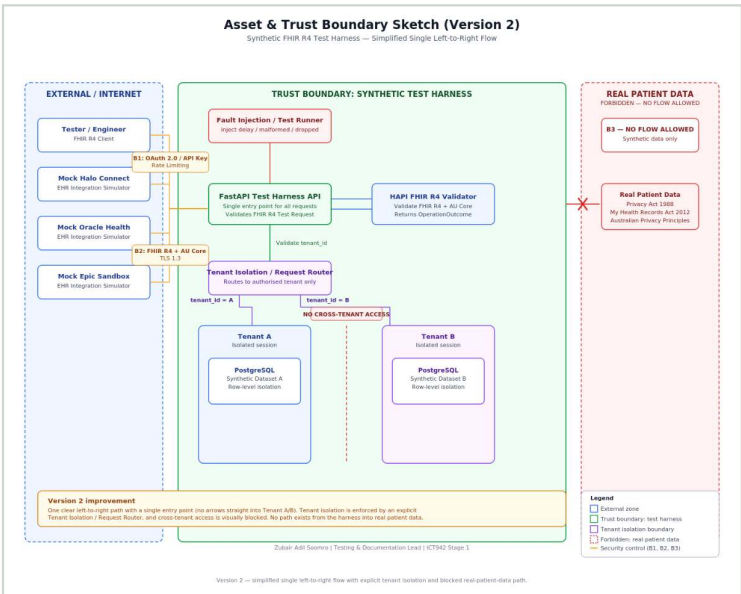
What you changed and why

I revised the figure by adding an internal tenant-isolation boundary within the Synthetic Test Harness and clearer arrows showing how data moves between the tester, FastAPI, FHIR validator, synthetic database and mock EMR systems. Version 1 showed the main trust boundary but did not explain how separate test tenants would be isolated or how information flowed between components. I added tenant-specific separation and security checks to reduce the risk of one synthetic test environment accessing another tenant's records. I also kept the B3 "No Flow Allowed" boundary to clearly separate the harness from real patient data.

Whose feedback changed it

Feedback from my group during the Week 6 stand-up, together with my own review of the first figure. The group agreed that the privacy and trust boundary needed to be visually clear rather than only explained in text.

W7-3-01 · 07 SEPT 2026

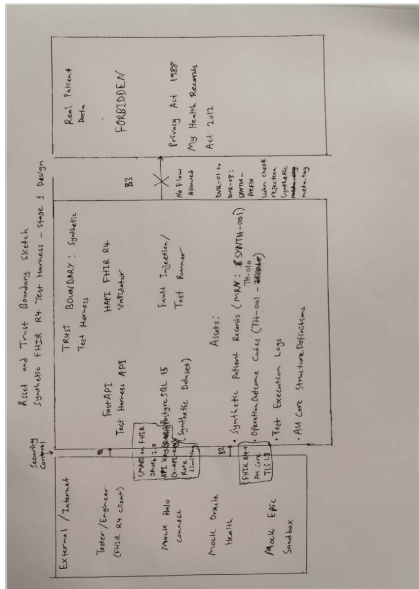


Section: 4.1 Proposed Solution Architecture – Asset and Trust Boundary Design. Also supports 4.3 Security Design and Control Mapping. · Task: Revise your figure – version 1 to version 2

My contribution: I revised my Asset & Trust Boundary Sketch to show the security flow more clearly. I added a single entry point through the FastAPI Test Harness API, a proposed Tenant Isolation / Request Router designed to validate tenant_id, separate Tenant A and Tenant B synthetic datasets, and a "NO CROSS-TENANT ACCESS" boundary. I also clarified the B1/B2 security controls and kept B3 as a blocked path so no connection to real patient data is permitted.

What it demonstrates: Version 1 identified the main trust zones and security controls, but it did not clearly show how external requests were routed or how different tenants would remain isolated. Version 2 shows that requests first pass through the security controls and FastAPI entry point, then the proposed tenant router would route requests only to the authorised synthetic tenant environment. Separate datasets and the "NO CROSS-TENANT ACCESS" boundary make tenant isolation explicit, while B3 clearly blocks any flow to real patient data. This demonstrates that I improved the figure based on security and privacy considerations rather than only changing its appearance. The earlier discussion also identified making the privacy/trust boundary structurally visible as an important improvement.

W7-3-02 · 07 SEPT 2026



Section: 4.1 Proposed Solution Architecture (trust boundary and security control mapping); also supports 4.3 Security Design and Control Mapping · Task: Draw the asset and trust boundary sketch showing the three security zones (External, Synthetic Harness, Forbidden Real Data), the data flow paths, and the security controls at each boundary.

My contribution: I designed and drew the sketch myself. I defined the three zones: External / Internet (Tester, Mock Halo Connect, Mock Oracle Health, Mock Epic Sandbox), the central Trust Boundary containing the four core components (FastAPI Test Harness API, HAPI FHIR R4 Validator, PostgreSQL 15 Synthetic Dataset, Fault Injection / Test Runner), and the Forbidden Real Patient Data zone. I mapped the security controls at each boundary — B1 (SMART on FHIR / OAuth 2.0, API Key, Rate Limiting), B2 (FHIR R4 + AU Core, TLS 1.3), and B3 (NO FLOW ALLOWED with DVR-01 to DVR-08). I also included the Assets box listing synthetic patient records, OperationOutcome codes, test logs, and AU Core StructureDefinitions.

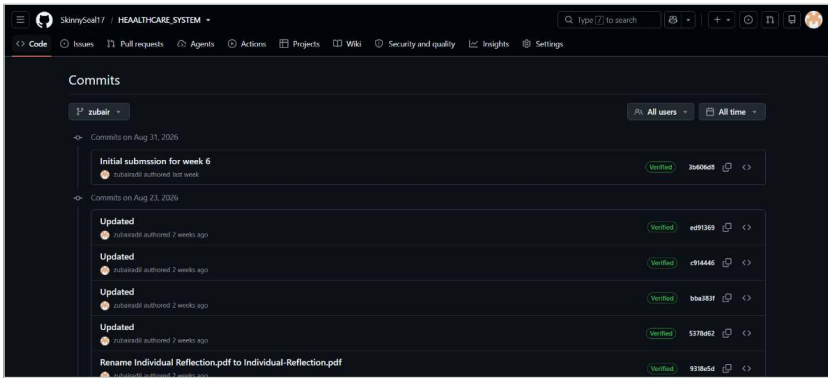
What it demonstrates: The first draft of the architecture diagram (Sonam's C4 container diagram) showed components and connectors but did not explicitly visualise the trust boundary separating synthetic data from real patient data. I revised my sketch after discussing with the group during the Week 6 stand-up, where we agreed that the Privacy Act 1988 compliance angle needed to be structurally visible, not just documented in text. The red forbidden zone with the crossed-out arrow makes the “no real data” rule impossible to miss, and the defence-in-depth controls at B1 and B2 show where the proposed architecture would apply identity validation, rate limiting and AU Core conformance checks before data moves.

W7-4 · Log a real GitHub action

Logged late · 6 min · 4/4 required parts · 07 Sept, 16:21

DATE	ISSUE / BRANCH / COMMIT	REFERENCE	WHAT YOU DID	LINK	OUTCOME
31 Aug 2026	Commit	3b606d8 – Initial submission for week 6	I committed my Week 6 project work to the zubair branch of the HEALTHCARE_SYSTEM GitHub repository using my GitHub account.	https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM/commit/3b606d899f307300f546f1ce8bd4eccc1e0df95	Commit was successfully added to the repository and is shown as Verified.

W7-4-01 · 07 SEPT 2026



Section: Section 7 – Individual Contribution / GitHub Activity. You can also note that it supports 5.1 Sprint Plan and 5.2 Proof of Concept Description if the committed Week 6 work relates to those sections.

- Task: Log a real GitHub action

My contribution: I used my GitHub account to commit my Week 6 project work to the zubair branch of the HEALTHCARE_SYSTEM repository. The commit is recorded under my username and provides a traceable record of my individual contribution to the team project.

What it demonstrates: This evidence demonstrates that I actively contributed to the shared GitHub repository rather than only participating in discussions. The commit history shows my username, branch, commit reference and verified status, providing a clear record of my project activity and supporting accountability for my individual work.

W7-5 • Record one technical decision

Logged late • 3 min • 4/4 required parts • 07 Sept, 16:24

Decision

Use a single FastAPI Test Harness API as the entry point, with a Tenant Isolation / Request Router validating tenant_id before requests can access Tenant A or Tenant B synthetic data.

Context — what forced the decision

Version 1 showed the external systems, trust boundary and synthetic database, but it did not clearly explain how requests would be separated if more than one tenant used the test harness. Direct connections from external systems to tenant data could make the security flow unclear and increase the risk of cross-tenant access. The revised design therefore needed one controlled entry point where authentication, FHIR validation and tenant identification could occur before a request reached any tenant-specific data.

Options considered

I considered three options: allowing each external system to connect directly to its tenant database, running a completely separate test-harness instance for every tenant, or using one shared FastAPI entry point with tenant-aware routing. Separate instances provide strong isolation but increase configuration and maintenance effort. Direct database access is simpler but weakens central security control. I selected the shared API and Tenant Isolation / Request Router because it provides a clear control point while still keeping Tenant A and Tenant B logically separated.

Consequence, including what gets harder

The benefit is that authentication, rate limiting, FHIR validation and tenant checks can be applied consistently before data is accessed. It also makes the trust boundary and data flow easier to audit. However, the router becomes a security-critical component. The design must correctly validate tenant_id, prevent cross-tenant queries and test isolation failures. Database rules such as row-level isolation also need careful configuration and testing, so the design is more complex than using a single unrestricted synthetic dataset.

W7-6 • Weekly reflection

Logged late • 0 min • 4/4 required parts • 07 Sept, 16:25

What did I personally complete this week?

This week I revised my Week 6 Asset & Trust Boundary Sketch into Version 2. I improved the design by adding a single FastAPI entry point, a Tenant Isolation / Request Router, separate Tenant A and Tenant B synthetic datasets, and a clear “NO CROSS-TENANT ACCESS” boundary. I also recorded a technical decision explaining why tenant-aware routing was selected and logged my GitHub activity as evidence of my contribution to the project.

What did I learn about the problem or the technology?

I learned that showing a trust boundary is not enough unless the diagram also explains how requests move through the system and how access is controlled. Tenant isolation is important because a shared test environment could create cross-tenant access risks if requests are not correctly routed. I also learned that using one controlled API entry point makes authentication, FHIR validation, tenant checking and auditing easier to understand and manage in the design.

What decision did I influence, and why?

I influenced the decision to use a single FastAPI Test Harness API followed by a Tenant Isolation / Request Router instead of allowing external systems to connect directly to tenant data. I supported this option because it creates one controlled point for authentication, rate limiting, FHIR validation and tenant_id checks. It also makes the security flow clearer and helps prevent Tenant A from accessing Tenant B's synthetic records.

What will I finish before the next class?

Before the next class, I will refine the Stage 1 architecture and supporting documentation so the tenant-isolation controls, FHIR validation flow and security boundaries are consistent across the report. I will also continue working on the Testing, Validation and Cybersecurity Evaluation section, including the planned failure-mode tests and privacy controls, and make sure my GitHub and portfolio evidence clearly matches the work I have completed.

6 of 6 entries logged · 4 minutes recorded · 3 evidence items · 100% complete

W8-1 · Show your component running

Logged late · 1 min · 4/4 required parts · 08 Sept, 16:12

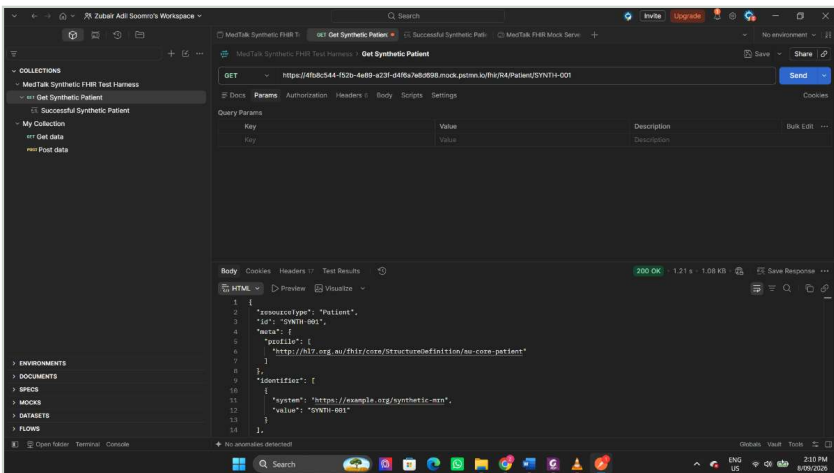
What you built or configured

I configured a synthetic FHIR R4 Patient mock endpoint in Postman as an initial proof-of-concept component of the Healthcare Integration & Clinical Terminology Test Harness. The endpoint responds to GET /fhir/R4/Patient/SYNTH-001 and returns a structured synthetic FHIR Patient resource. I created the request, configured the example response, created the Postman mock server and successfully executed the request. The endpoint returned HTTP 200 OK together with the expected FHIR JSON. This demonstrates a working request-response component using synthetic data only and provides a small, verifiable foundation for later validation, terminology-mapping and fault-injection testing.

How to run or reproduce it

Open the MedTalk Synthetic FHIR Test Harness collection in Postman and select the synthetic Patient request. Confirm that the request uses the configured Postman mock-server URL ending in /fhir/R4/Patient/SYNTH-001, then click Send. A successful execution returns HTTP 200 OK and a JSON response containing resourceType: Patient and id: SYNTH-001. The same request can be sent repeatedly using the saved Postman configuration.

W8-1-01 · 08 SEPT 2026



Section: 5.2 Proof of Concept Description · Task: Show my component running

My contribution: I configured and tested a synthetic FHIR R4 Patient mock endpoint in Postman as part of the Healthcare Integration & Clinical Terminology Test Harness. I created a GET request for /fhir/R4/Patient/SYNTH-001, configured a synthetic FHIR Patient response, and used a Postman mock server to execute the request. I then verified that the endpoint returned the expected HTTP 200 OK response and FHIR Patient JSON using synthetic data only.

What it demonstrates: The screenshot demonstrates that the synthetic FHIR Patient endpoint is configured and running successfully. It shows a GET request being sent to the mock FHIR R4 endpoint and a successful 200 OK response containing a synthetic Patient resource. This provides practical evidence that an initial component of the planned synthetic test harness has moved from the Stage 1 design into a working proof-of-concept configuration demonstrating the feasibility of the planned test harness.

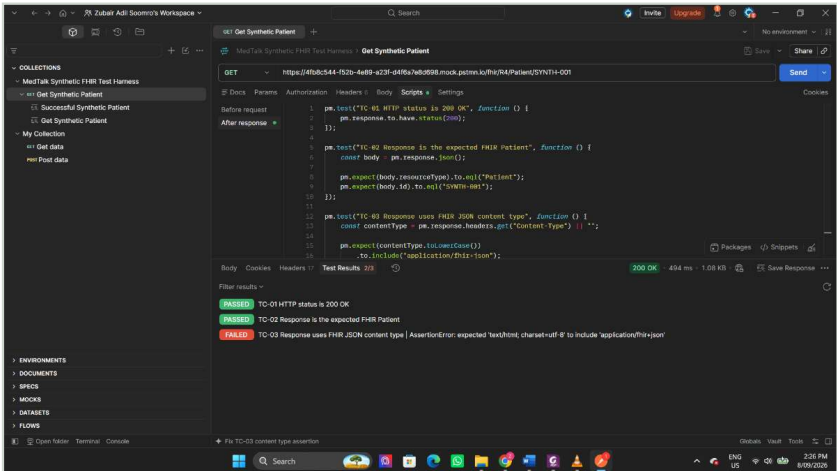
W8-2 · Test it and record the results

Logged late · 1 min · 4/4 required parts · 08 Sept, 14:27

TEST ID	WHAT IS BEING TESTED	INPUT / STEPS	EXPECTED	ACTUAL	PASS / FAIL
TC-01	Successful HTTP response from the synthetic FHIR Patient endpoint.	Send a GET request to /fhir/R4/Patient/SYNTH-001 using the configured Postman mock server.	The endpoint should return HTTP 200 OK.	HTTP 200 OK was returned.	Pass

TEST ID	WHAT IS BEING TESTED	INPUT / STEPS	EXPECTED	ACTUAL	PASS / FAIL
TC-02	Correct FHIR Patient resource and synthetic patient identifier	Send the GET request and inspect the JSON response. Verify that resourceType equals Patient and id equals SYNTH-001.	The response should contain resourceType: Patient and the synthetic identifier SYNTH-001.	The response contained resourceType: Patient and id: SYNTH-001.	Pass
TC-03	FHIR-specific response Content-Type configuration.	Send the Patient request and check the response Content-Type header using the Postman test script.	The response should use application/fhir+json to identify the payload as FHIR JSON.	The endpoint returned the Patient data successfully, but the response did not contain application/fhir+json as its Content-Type.	Fail

W8-2-01 · 08 SEPT 2026



Section: 5.3 Testing, Validation and Cybersecurity Evaluation · Task: Test it and record the results

My contribution: I configured and executed three Postman tests against the synthetic FHIR R4 Patient mock endpoint. I tested the HTTP response status, validated the returned FHIR Patient resource and synthetic identifier, and checked whether the response used the FHIR-specific JSON Content-Type. I recorded the expected and actual outcomes and identified a response-header configuration issue for further improvement.

What it demonstrates: The evidence demonstrates that the synthetic FHIR Patient component was tested against clearly defined expected outcomes rather than only being shown running. The functional tests confirmed that the endpoint returns HTTP 200 and the expected synthetic Patient resource. The Content-Type test identified a configuration gap, which has been documented as a failed test for correction and regression testing.

W8-3 · Map your work to a security framework

Logged late · 0 min · 3/3 required parts · 08 Sept, 16:02

Framework used

NIST Cybersecurity Framework 2.0

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
PR.DS	Data Security	The Postman FHIR Patient mock endpoint uses synthetic patient information only, such as SYNTH-001, rather than real clinical data. This reduces the risk of exposing real patient information during testing. However, the current PoC evidence does not demonstrate encryption at rest, formal secure storage or data-retention controls.	Partial

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
PR.AA	Identity Management, Authentication and Access Control	The project requires the test environment to be restricted to authorised users, but the current Postman mock endpoint does not yet implement authentication, role-based access control or OAuth. These controls are planned for later development of the test harness.	Planned
DE.CM	Continuous Monitoring	The current Postman PoC provides manual test evidence by checking HTTP status, FHIR response structure and response headers. This demonstrated that defined checks can identify configuration issues, including the failed FHIR Content-Type test. However, continuous logging, event monitoring, security alerts and automated monitoring are not implemented in the current PoC.	Planned

W8-4 · Ethics, privacy, legal and AI governance

Logged late · 0 min · 3/3 required parts · 08 Sept, 14:54

Personal data your solution touches, and the obligation that applies

The current PoC does not use real personal or patient data. The Postman mock endpoint uses a synthetic FHIR Patient record containing a made-up identifier, name, gender and date of birth. In a production healthcare environment, these fields would represent personal and health information and would require protection under the Privacy Act 1988 (Cth) and the Australian Privacy Principles. For this project, the main privacy control is strict use of synthetic data so that no real patient information enters the test harness. My Health Record connectivity is not implemented in the PoC, so My Health Record data is not currently accessed or stored.

AI use in this project and how it was governed

AI tools were used to support learning and development activities, including explaining FHIR and Postman concepts, suggesting test scenarios, and helping structure technical documentation. AI output was not treated as evidence of implementation. Each technical step was performed and verified manually in Postman, and the pass/fail results recorded in Week 8 came from the actual test execution. AI-assisted suggestions were reviewed before use, and significant assistance was documented so that the work remained traceable. Any technical configuration or test result included in the project must be understood and explainable by the team.

One ethical or legal risk you cannot fully mitigate

A risk that cannot be completely eliminated is incorrect clinical terminology mapping between SNOMED CT-AU, ICD-10-AM and MBS codes. Even when synthetic records and predefined test cases pass, mappings may be ambiguous, change between terminology releases, or behave differently in a real clinical system. An incorrect mapping could eventually contribute to inaccurate clinical documentation or billing if similar logic were used in production. The test harness can reduce this risk through edge-case testing, version-controlled mappings and manual review, but it cannot guarantee that every real-world clinical context will be interpreted correctly.

W8-5 · Own a slide

Logged late · 0 min · 5/5 required parts · 08 Sept, 16:40

Slide title and number you are responsible for

Slide 9 – Proof of Concept, Testing and Cybersecurity Evaluation

The three points you will make

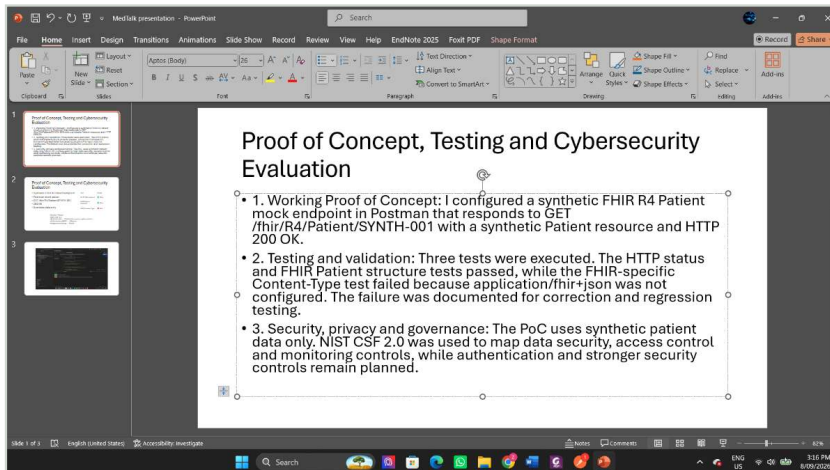
1. Working Proof of Concept: I configured a synthetic FHIR R4 Patient mock endpoint in Postman that responds to GET /fhir/R4/Patient/SYNTH-001 with a synthetic Patient resource and HTTP 200 OK.
2. Testing and validation: Three tests were executed. The HTTP status and FHIR Patient structure tests passed, while the FHIR-specific Content-Type test failed because application/fhir+json was not configured. The failure was documented for correction and regression testing.
3. Security, privacy and governance: The PoC uses synthetic patient data only. NIST CSF 2.0 was used to map data security, access control and monitoring controls, while authentication and stronger security controls remain planned.

A question you expect from the audience, and your answer

Question: Why is a failed test included in the demonstration instead of fixing it before presenting?

Answer: The failed Content-Type test demonstrates that the test process can identify a real configuration gap rather than simply showing successful results. The endpoint functions correctly and returns the expected synthetic FHIR Patient data, but the response should use the FHIR-specific application/fhir+json content type. The issue has therefore been documented for correction and later regression testing. This provides stronger evidence of systematic validation than reporting only successful tests.

W8-5-01 · 08 SEPT 2026



Section: 5. Proof of Concept and Evaluation · Task: Own a slide

My contribution: I prepared the presentation slide covering the Proof of Concept, testing results, cybersecurity evaluation, privacy considerations and identified improvement areas. The slide summarises evidence from my Postman FHIR Patient mock endpoint and the Week 8 test results, including the documented failed Content-Type test.

What it demonstrates: The slide demonstrates my individual contribution to presenting the implementation and evaluation work assigned through HITH-45 and HITH-46. It shows the working synthetic FHIR proof of concept, actual test outcomes, and an honest assessment of which cybersecurity controls are partially implemented or still planned.

W8-6 · Limitations, peer cross-check and final reflection

Logged late · 1 min · 7/7 required parts · 08 Sept, 15:36

Honest limitations of your contribution

My current contribution is an early proof of concept rather than a complete test harness. The Postman implementation currently demonstrates only a synthetic FHIR Patient endpoint and a small set of validation tests. It does not yet implement authentication, role-based access control, automated regression execution, terminology mapping, or fault-injection scenarios such as malformed data, delayed responses and dropped connections. One test also identified that the FHIR-specific application/fhir+json Content-Type was not configured. These limitations are documented so that the component can be expanded and retested during the next development stage.

Which group member's evidence did you review today?

Rikesh Aryal

One specific thing that was strong and one thing that needs work in their evidence

Rikesh's evidence was strong because it clearly connected the technical work to the FHIR integration requirements and showed how the component related to the wider MedTalk Connect test-harness architecture. One area that could be improved is the amount of execution evidence shown. The submission would be stronger if the screenshot or tool output made the actual result more visible, such as the request, returned status, expected result and whether the test passed or failed.

What did I personally complete this week?

This week I configured and demonstrated a synthetic FHIR R4 Patient mock endpoint in Postman. I executed three tests covering the HTTP response, FHIR Patient structure and response Content-Type, recording two passes and one documented failure. I also mapped the PoC to NIST Cybersecurity Framework 2.0 controls, assessed privacy, ethics, legal

and AI-governance considerations, and prepared my presentation slide covering the proof of concept and testing results.

What did I learn about the problem or the technology?

I learned how Postman mock servers can be used to simulate a FHIR API before a complete backend has been developed. I also learned that a successful 200 OK response alone does not prove FHIR conformance. Response structure, identifiers, headers and other expected behaviours need separate validation. The failed Content-Type test showed why clearly defined expected results are important and why documented failures can provide useful evidence for improving the test harness.

What decision did I influence, and why?

I supported keeping the Week 8 proof of concept small and verifiable by focusing first on a synthetic FHIR Patient endpoint rather than attempting to implement the entire test harness at once. This allowed us to demonstrate a working component, define measurable tests and identify a real configuration issue. The approach also reduces scope risk and provides a stable foundation for adding terminology mapping, negative testing and other FHIR resources later.

What will I finish before the next class?

Before the next class, I will finalise, integrate and prepare for submission my assigned report contributions: HITH-37 Sprint Plan, HITH-45 Proof of Concept Description, HITH-46 Testing, Validation and Cybersecurity Evaluation including ethics and privacy, HITH-47 Conclusion and Recommendations, and HITH-48 Executive Summary. I will incorporate the Week 8 Postman proof-of-concept evidence and test results into Sections 5.2 and 5.3, check consistency with the team report, and update each Jira task with the final evidence and completion status.

Literature and evidence tracker

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
6	Walonoski, J., Scanlon, R., Dowling, C., Hyland, M., Ettema, R., & Posnack, S. (2018). Validation and testing of Fast Healthcare Interoperability Resources standards compliance: Data analysis. JMIR Medical Informatics, 6(4), e10870.	Google Scholar, Google	Crucible and Touchstone are independent FHIR validation tools. Their usage data shows a statistically significant positive correlation between frequent testing and vendor server compliance ($r^2 = .80$, $P < .005$, $R^2 = .26$). "Touchstone makes perfect" — developers who test more produce more compliant FHIR implementations.	Feeds Section 3.2 (Technology and Market Survey) by proving that existing FHIR testing tools exist but are generic; our synthetic harness is differentiated because it is Australian-context-specific (AU Core, SNOMED CT-AU, ICD-10-AM, MBS) and deliberately isolates real patient data.
6	Nguyen, A. N., Truran, D., Kemp, M., Koopman, B., Conlan, D., O'Dwyer, J., et al. (2018). Computer-assisted diagnostic coding: effectiveness of an NLP-based approach using SNOMED CT to ICD-10 mappings. AMIA Annual Symposium Proceedings, 2018, 807–816.	Google Scholar, Google	SNOMED CT to ICD-10-AM mapping via NLP achieved only 54.1% sensitivity and 70.2% positive predictive value. Even simple rule-based approaches struggle with edge cases (e.g., iron deficiency anemia at 3% sensitivity). Manual coding validation itself only reaches 76.3% sensitivity, proving this is a hard problem.	Feeds Section 3.2 and justifies our project: if automated SNOMED CT to ICD-10-AM mapping is error-prone in production, a repeatable synthetic test harness is essential to catch mapping failures before they reach live EMRs. Also cited in Section 4.2 to justify why we need dedicated mapping validation.
6	HL7 Australia. (2023). AU Core FHIR Implementation Guide (v0.2.0-preview).	Google Scholar, Google	Defines the Australian FHIR profiles (Patient, Encounter, Observation, etc.) that our test harness must validate against. Specifies mandatory identifiers (IHI, Medicare Provider Number) and RESTful interactions. This is the current Australian baseline, not generic FHIR R4.	Feeds Section 3.2 by proving AU Core is the compliance baseline our harness must test against; referenced in Section 4.1 (architecture) and Section 4.3 (control mapping) to show our solution aligns with the Australian Digital Health Agency's interoperability roadmap.
7	HAPI FHIR (JPA Server + Validation Engine) (University Health Network / HAPI FHIR Project. Open source, Apache 2.0.)	hapifhir.io; Nirmittee (2026); GitHub — hapifhir/hapi-fhir-jpaserver-s-tarter	Full FHIR R4/R5 compliance; passes HL7 conformance test suite; programmable validation against published StructureDefinitions (including AU Core); Docker-deployable; supports custom interceptors and \$export; 4,000+ GitHub stars with active community.	No built-in security implementation, no enterprise logging, and no built-in de-identification out of the box — all require custom configuration. Self-managed, so production deployments need significant DevOps overhead. Search performance degrades beyond ~50M resources without Elasticsearch tuning. US-centric defaults; AU Core profiling must be manually configured.
7	Touchstone (AEGIS.net. IaaS / TaaS open-access solution (commercial-hosted, free registration).)	Medigy; help.edifecsfecloud.com; AEGIS Touchstone site	Automated, internet-based FHIR interoperability testing against HL7 FHIR specifications and standards. Thousands of conformance tests using the FHIR TestScript resource. Self-registration of accounts and organisations. Peer-to-peer testing against other FHIR server/client implementations. Drill-down to individual operations and assertions.	Internet-based SaaS — requires live connectivity and exposes test systems externally. Generic FHIR conformance testing with no Australian-context specificity: no AU Core profiles, no SNOMED CT-AU, no ICD-10-AM or MBS billing code validation. Not designed for synthetic data isolation or clinical-safety fault injection. No terminology mapping validation between code systems.

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
7	Synthea (MITRE Corporation. Open source, Apache 2.0.)	GitHub – syntheticea	Synthetic patient population lth/synthea; Walonoski simulator generating clinically et al. (JAMIA 2018); realistic records from birth to Nirmitee (2026); Veria Exports FHIR R4 Bundles, C-CDA, and CSV. Uses standard terminologies (SNOMED CT, LOINC, RxNorm, ICD-10-CM). Temporal consistency across encounters, conditions, and observations. US Core conformance. Free, no PHI, no BAA required.	US-centric by default – demographics, insurance details, and treatment guidelines are American. Australian adaptations (Synthea International) exist but are community-maintained and incomplete. Probabilistic generation means no scenario control – cannot reliably generate a specific 55-year-old diabetic with hypertension for a targeted test case. No vendor-specific FHIR data shapes (Epic, Cerner, Halo Connect all emit differently). No built-in AU Core profile validation. Requires post-processing to enforce Australian coding rules.
8	NIST Cybersecurity Framework 2.0 PR.DS	framework document	Data Security	The Postman FHIR Patient mock endpoint uses synthetic patient information only, such as SYNTH-001, rather than real clinical data. This reduces the risk of exposing real patient information during testing. However, the current PoC evidence does not demonstrate encryption at rest, formal secure storage or data-retention controls.
8	NIST Cybersecurity Framework 2.0 PR.AA	framework document	Identity Management, Authentication and Access Control	The project requires the test environment to be restricted to authorised users, but the current Postman mock endpoint does not yet implement authentication, role-based access control or OAuth. These controls are planned for later development of the test harness.
8	NIST Cybersecurity Framework 2.0 DE.CM	framework document	Continuous Monitoring	The current Postman PoC provides manual test evidence by checking HTTP status, FHIR response structure and response headers. This demonstrated that defined checks can identify configuration issues, including the failed FHIR Content-Type test. However, continuous logging, event monitoring, security alerts and automated monitoring are not implemented in the current PoC.

GitHub and project-management activity

DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
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DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
31 Aug 2026	GitHub	Commit 3b606d8 - Initial submission for week 6	I committed my Week 6 project work to the zubair branch of the HEALTHCARE_SYSTEM GitHub repository using my GitHub account.	https://github.com/SkitchSeal17/HEALTHCARE_SYSTEM/commit/3b606d899f307300f546f1ce8bd4eccc1e0df95	Commit was successfully added to the repository and is shown as Verified.
Wk6	Board	HITH-37 Sprint Plan — design the two-spr	Created or took ownership		
Wk6	Board	HITH-45 Proof of Concept Description — d	Created or took ownership		
Wk6	Board	HITH-46 Testing, Validation, and Cyberse	Created or took ownership		
Wk6	Board	HITH-47 Conclusion and Recommendations —	Created or took ownership		
Wk6	Board	HITH-48 Executive Summary — compose the	Created or took ownership		

Evidence index

NO .	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W6-2-01	03 Sept 2026	5.1 Sprint Plan	Literature search log	The sprint plan de-risks the build by sequencing the error-handler first—every downstream component depends on
W6-2-02	03 Sept 2026	5.2 Proof of Concept Description	Literature search log	The PoC shows iterative build discipline—starting with a minimal request-response loop rather than full connec
W6-2-03	03 Sept 2026	5.3 Testing, Validation, and Cybersecurity Evaluation	Literature search log	The test design directly addresses the clinical-safety risk: Nguyen et al. (2018) proves S N O M E D ! ' I C D m a p p i n g
W6-3-01	03 Sept 2026	Sections 5.1, 5.2, 5.3, 6, and Executive Summary	Claim your tasks on the board	The Jira screenshot shows all five tasks (HITH-37, HITH-45, HITH-46, HITH-47, HITH-48) assigned to me, proving
W6-4-01	07 Sept 2026	4.1 Proposed Solution Architecture (trust boundary and security control mapping); also supports 4.3 Security Design and Control Mapping	Draw the first figure	The first draft of the architecture diagram (Sonam's C4 container diagram) showed components and connectors bu
W7-1-01	07 Sept 2026	3.2 Technology and Market Survey; 3.3 Comparative Analysis of Existing Solutions	Compare three existing tools	The screenshot proves I consulted primary vendor documentation, not secondary summaries, to assess HAPI FHIR's
W7-3-01	07 Sept 2026	4.1 Proposed Solution Architecture — Asset and Trust Boundary Design. Also supports 4.3 Security Design and Control Mapping.	Revise your figure — version 1 to version 2	Version 1 identified the main trust zones and security controls, but it did not clearly show how external requ

NO.	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W7-3-02	07 Sept 2026	4.1 Proposed Solution Architecture (trust boundary and security control mapping); also supports 4.3 Security Design and Control Mapping	Revise your figure – version 2	The first draft of the architecture diagram (Sonam's C4 container diagram) showed components and connectors bu
W7-4-01	07 Sept 2026	Section 7 – Individual Contribution / GitHub Activity. You can also note that it supports 5.1 Sprint Plan and 5.2 Proof of Concept Description if the committed Week 6 work relates to those sections.	Log a real GitHub action	This evidence demonstrates that I actively contributed to the shared GitHub repository rather than only partic
W8-1-01	08 Sept 2026	5.2 Proof of Concept Description	Show your component running	The screenshot demonstrates that the synthetic FHIR Patient endpoint is configured and running successfully. I
W8-2-01	08 Sept 2026	5.3 Testing, Validation and Cybersecurity Evaluation	Test it and record the results	The evidence demonstrates that the synthetic FHIR Patient component was tested against clearly defined expecte
W8-5-01	08 Sept 2026	5. Proof of Concept and Evaluation	Own a slide	The slide demonstrates my individual contribution to presenting the implementation and evaluation work assigne

VERIFICATION

Time log

Each entry below was recorded automatically while the task was open in the portfolio tool. The timer pauses after three minutes without activity.

TASK	SESSION START	DURATION	CONTEXT
S-1	24 Aug, 10:22	00:30	In class
S-2	24 Aug, 10:22	00:05	In class
S-1	31 Aug, 11:01	00:06	Outside class
S-1	31 Aug, 11:09	00:26	Outside class
S-1	31 Aug, 11:09	00:30	Outside class
S-1	31 Aug, 11:10	00:30	Outside class
S-1	31 Aug, 11:10	00:17	Outside class
S-1	31 Aug, 11:11	00:30	Outside class
S-1	31 Aug, 11:12	00:09	Outside class
S-1	31 Aug, 11:12	00:12	Outside class
S-1	31 Aug, 11:13	00:05	Outside class
S-1	31 Aug, 11:14	00:30	Outside class
S-1	31 Aug, 11:14	00:09	Outside class
S-1	31 Aug, 11:15	00:10	Outside class
S-2	31 Aug, 11:15	00:30	Outside class
S-2	31 Aug, 11:15	00:11	Outside class
S-1	31 Aug, 11:16	00:12	Outside class
S-1	31 Aug, 11:16	00:10	Outside class
S-1	31 Aug, 11:16	00:05	Outside class
S-1	31 Aug, 11:16	00:06	Outside class
S-2	31 Aug, 11:16	00:05	Outside class
S-2	31 Aug, 11:16	00:09	Outside class
S-2	31 Aug, 11:16	00:05	Outside class
S-2	31 Aug, 11:16	00:07	Outside class
S-1	31 Aug, 11:17	00:08	Outside class
S-1	31 Aug, 11:17	00:26	Outside class
S-2	31 Aug, 11:17	00:07	Outside class
S-2	31 Aug, 11:17	00:12	Outside class
S-1	31 Aug, 11:19	00:30	Outside class
S-1	31 Aug, 11:20	00:09	Outside class
S-1	31 Aug, 11:20	00:08	Outside class
S-2	31 Aug, 11:20	00:11	Outside class
S-2	31 Aug, 11:25	00:05	Outside class

TASK	SESSION START	DURATION	CONTEXT
S-2	31 Aug, 11:25	00:13	Outside class
S-2	31 Aug, 11:26	00:05	Outside class
S-2	31 Aug, 11:26	00:05	Outside class
S-2	31 Aug, 11:26	00:11	Outside class
S-2	31 Aug, 11:28	00:10	Outside class
S-2	31 Aug, 11:28	00:05	Outside class
S-2	31 Aug, 11:29	00:30	Outside class
S-2	31 Aug, 11:30	00:31	Outside class
S-2	31 Aug, 11:30	00:31	Outside class
S-2	31 Aug, 11:31	00:30	Outside class
S-2	31 Aug, 11:31	00:06	Outside class
S-2	31 Aug, 12:06	00:30	Outside class
S-2	31 Aug, 12:07	00:31	Outside class
S-2	31 Aug, 12:07	00:31	Outside class
S-2	31 Aug, 12:08	00:28	Outside class
S-2	31 Aug, 12:09	00:30	Outside class
S-2	31 Aug, 12:09	00:31	Outside class
S-2	31 Aug, 12:10	00:31	Outside class
S-2	31 Aug, 12:10	00:31	Outside class
S-2	31 Aug, 12:11	00:31	Outside class
S-2	31 Aug, 12:11	00:31	Outside class
S-2	31 Aug, 12:12	00:22	Outside class
S-2	31 Aug, 12:15	00:30	Outside class
S-2	31 Aug, 12:16	00:31	Outside class
S-2	31 Aug, 12:16	00:28	Outside class
S-2	31 Aug, 12:24	00:30	Outside class
S-2	31 Aug, 12:25	00:31	Outside class
S-2	31 Aug, 12:25	00:30	Outside class
S-2	31 Aug, 12:26	00:31	Outside class
S-2	31 Aug, 12:26	00:31	Outside class
S-2	31 Aug, 12:27	00:30	Outside class
S-2	31 Aug, 12:28	00:30	Outside class
S-2	31 Aug, 12:28	00:31	Outside class
S-2	31 Aug, 12:29	00:30	Outside class
S-2	31 Aug, 12:29	00:31	Outside class
S-2	31 Aug, 12:30	00:23	Outside class
S-2	31 Aug, 15:02	00:30	Outside class

TASK	SESSION START	DURATION	CONTEXT
S-2	31 Aug, 15:02	00:30	Outside class
S-2	31 Aug, 15:03	00:31	Outside class
S-2	31 Aug, 15:03	00:31	Outside class
S-2	31 Aug, 15:04	00:30	Outside class
S-2	31 Aug, 15:04	00:19	Outside class
W6-1	31 Aug, 15:05	00:30	Outside class
W6-1	31 Aug, 15:05	00:10	Outside class
W6-1	31 Aug, 15:06	00:16	Outside class
W7-1	31 Aug, 15:06	00:11	Outside class
W7-2	31 Aug, 15:06	00:08	Outside class
W6-1	31 Aug, 15:07	00:28	Outside class
W6-1	31 Aug, 15:08	00:06	Outside class
W6-1	31 Aug, 15:08	00:12	Outside class
W6-1	31 Aug, 15:09	00:18	Outside class
W6-1	31 Aug, 15:09	00:30	Outside class
W6-1	31 Aug, 15:10	00:24	Outside class
W6-1	31 Aug, 15:10	00:31	Outside class
W6-1	31 Aug, 15:11	00:31	Outside class
W6-1	31 Aug, 15:11	00:31	Outside class
W6-1	31 Aug, 15:12	00:30	Outside class
W6-1	31 Aug, 15:12	00:31	Outside class
W6-1	31 Aug, 15:13	00:10	Outside class
W6-1	31 Aug, 15:13	00:30	Outside class
W6-1	31 Aug, 15:13	00:10	Outside class
W6-1	31 Aug, 15:14	00:30	Outside class
W6-1	31 Aug, 15:15	00:30	Outside class
W6-1	31 Aug, 15:15	00:30	Outside class
W6-1	31 Aug, 15:16	00:31	Outside class
W6-1	31 Aug, 15:16	00:31	Outside class
W6-1	31 Aug, 15:17	00:25	Outside class
W6-1	31 Aug, 15:19	00:30	Outside class
W6-1	31 Aug, 15:20	00:31	Outside class
W6-1	31 Aug, 15:20	00:31	Outside class
W6-1	31 Aug, 15:21	00:31	Outside class
W6-1	31 Aug, 15:21	00:30	Outside class
W6-1	31 Aug, 15:22	00:31	Outside class
W6-1	31 Aug, 15:22	00:31	Outside class

TASK	SESSION START	DURATION	CONTEXT
W6-1	31 Aug, 15:23	00:31	Outside class
W6-1	31 Aug, 15:23	00:31	Outside class
W6-1	31 Aug, 15:24	00:23	Outside class
W6-1	31 Aug, 15:28	00:11	Outside class
W6-2	31 Aug, 15:29	00:30	Outside class
W6-2	31 Aug, 15:29	00:07	Outside class
W6-3	31 Aug, 15:29	00:19	Outside class
W6-2	31 Aug, 15:30	00:07	Outside class
W6-4	31 Aug, 15:30	00:30	Outside class
W6-4	31 Aug, 15:30	00:16	Outside class
W6-2	31 Aug, 15:31	00:30	Outside class
W6-2	31 Aug, 15:31	00:31	Outside class
W6-2	31 Aug, 15:32	00:31	Outside class

Weekly individual reflection

Week 6

What did I personally complete this week?

I completed the initial drafts of the Sprint Plan (HITH-37) and Proof of Concept Description (HITH-45). For the sprint plan, I designed a two-sprint Agile schedule for Stage 2, assigned design tasks to each team member, and defined the Definition of Done requiring peer-reviewed documents and mentor-reviewable specifications. For the PoC description, I drafted the narrative specifying the three-service Docker Compose stack, the single GET /fhir/R4/Patient/SYNTH-001 endpoint, and the four architectural assumptions being de-risked. I explicitly scoped out HL7 v2.4 MLLP, SNOMED mapping, rate limiting, and fault injection for future Stage 2, ensuring the Week 6 deliverable remained a design document only.

What did I learn about the problem or the technology?

I learned that the primary technical risk is not tool availability but healthcare semantics. Docker Compose, FastAPI, HAPI FHIR, and PostgreSQL are all mature and well-documented; the real challenge is correctly implementing AU Core profile validation, HL7 v2 segment parsing, and SNOMED CT-AU to ICD-10-AM mapping accuracy. Walonoski et al. (2018) showed that frequent FHIR testing improves compliance, which reinforced why our synthetic harness must be repeatable. I also learned that synthetic data generation carries its own privacy risk — Charity flagged that valid-looking IHI's could accidentally be generated, so I added DVR-02 and TC-MAL-03 to the design.

What decision did I influence, and why?

I influenced the decision to split the original single-sprint plan into two sprints. Rikesh flagged during our stand-up that specifying container networking for HAPI FHIR and PostgreSQL across mixed student environments needed dedicated design time before adding connector complexity. I argued that healthcare semantics — AU Core profiling and HL7 v2 parsing — is the primary risk, not tooling, so Sprint 1 should validate the architecture assumptions before Sprint 2 adds MLLP, fault injection, and mapping logic. The team agreed, and the mentor approved the revised timeline. This decision de-risked the Stage 1 deliverables and prevented scope creep.

What will I finish before the next class?

Before the next class, I will complete the Testing, Validation, and Cybersecurity Evaluation design (HITH-46). This includes finalising the seven failure-mode test cases (TC-SLOW, TC-MAL, TC-DROP, TC-UNAV), locking the OperationOutcome error code schema (TH-001–TH-010), and completing the Threat! Asset! Vulnerability also draft the Conclusion and Recommendations (HITH-47), synthesising the four feasibility dimensions and producing the priority-ordered next-steps table with mentor sign-off gates. I need to confirm the 50-record synthetic dataset freeze with Charity and schedule the Week 7 security review with Atif Nisar.

Week 7

What did I personally complete this week?

This week I revised my Week 6 Asset & Trust Boundary Sketch into Version 2. I improved the design by adding a single FastAPI entry point, a Tenant Isolation / Request Router, separate Tenant A and Tenant B synthetic datasets, and a clear “NO CROSS-TENANT ACCESS” boundary. I also recorded a technical decision explaining why tenant-aware routing was selected and logged my GitHub activity as evidence of my contribution to the project.

What did I learn about the problem or the technology?

I learned that showing a trust boundary is not enough unless the diagram also explains how requests move through the system and how access is controlled. Tenant isolation is important because a shared test environment could create cross-tenant access risks if requests are not correctly routed. I also learned that using one controlled API entry point makes authentication, FHIR validation, tenant checking and auditing easier to understand and manage in the design.

What decision did I influence, and why?

I influenced the decision to use a single FastAPI Test Harness API followed by a Tenant Isolation / Request Router instead of allowing external systems to connect directly to tenant data. I supported this option because it creates one controlled point for authentication, rate limiting, FHIR validation and tenant_id checks. It also makes the security flow clearer and helps prevent Tenant A from accessing Tenant B's synthetic records.

What will I finish before the next class?

Before the next class, I will refine the Stage 1 architecture and supporting documentation so the tenant-isolation controls, FHIR validation flow and security boundaries are consistent across the report. I will also continue working on the Testing, Validation and Cybersecurity Evaluation section, including the planned failure-mode tests and privacy controls, and make

sure my GitHub and portfolio evidence clearly matches the work I have completed.

Week 8

What did I personally complete this week?

This week I configured and demonstrated a synthetic FHIR R4 Patient mock endpoint in Postman. I executed three tests covering the HTTP response, FHIR Patient structure and response Content-Type, recording two passes and one documented failure. I also mapped the PoC to NIST Cybersecurity Framework 2.0 controls, assessed privacy, ethics, legal and AI-governance considerations, and prepared my presentation slide covering the proof of concept and testing results.

What did I learn about the problem or the technology?

I learned how Postman mock servers can be used to simulate a FHIR API before a complete backend has been developed. I also learned that a successful 200 OK response alone does not prove FHIR conformance. Response structure, identifiers, headers and other expected behaviours need separate validation. The failed Content-Type test showed why clearly defined expected results are important and why documented failures can provide useful evidence for improving the test harness.

What decision did I influence, and why?

I supported keeping the Week 8 proof of concept small and verifiable by focusing first on a synthetic FHIR Patient endpoint rather than attempting to implement the entire test harness at once. This allowed us to demonstrate a working component, define measurable tests and identify a real configuration issue. The approach also reduces scope risk and provides a stable foundation for adding terminology mapping, negative testing and other FHIR resources later.

What will I finish before the next class?

Before the next class, I will finalise, integrate and prepare for submission my assigned report contributions: HITH-37 Sprint Plan, HITH-45 Proof of Concept Description, HITH-46 Testing, Validation and Cybersecurity Evaluation including ethics and privacy, HITH-47 Conclusion and Recommendations, and HITH-48 Executive Summary. I will incorporate the Week 8 Postman proof-of-concept evidence and test results into Sections 5.2 and 5.3, check consistency with the team report, and update each Jira task with the final evidence and completion status.

Declaration

I confirm that the work recorded in this portfolio is my own individual contribution to the group project. Screenshots and links refer to work I personally performed. Where AI tools were used, this is disclosed in the Week 8 governance entry.

Signed: _____

Date: _____